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Energy Conversion

Assignment 6

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1 Problem 1

$$n_s = \frac{120f_e}{p} \implies 1000 = \frac{120 \times 50}{p} \implies p = 6 \text{ poles}$$

- **Stator magnetic field speed:** Equal to the synchronous speed, $n_s = 1000$ rpm.
- **Rotor magnetic field speed:** The rotor magnetic field revolves at synchronous speed relative to the stator, so it is also 1000 rpm. Relative to the rotor itself, it revolves at the slip speed ($n_{slip} = n_s - n_m$).

At No-Load:

$$s_{nl} = \frac{n_s - n_{nl}}{n_s} = \frac{1000 - 980}{1000} = 0.02$$

$$f_{r,nl} = s_{nl}f_e = 0.02 \times 50 = 1.0 \text{ Hz}$$

At Full-Load:

$$s_{fl} = \frac{n_s - n_{fl}}{n_s} = 0.07$$

$$f_{r,fl} = s_{fl}f_e = 3.5 \text{ Hz}$$

$$SR = \frac{n_{nl} - n_{fl}}{n_{fl}} \times 100\% \approx 5.38\%$$

2 Problem 2

For a Y-connected motor, the phase voltage is:

$$V_\phi = \frac{V_L}{\sqrt{3}} = \frac{400}{\sqrt{3}} \approx 230.94 \text{ V}$$

The synchronous angular speed is:

$$\omega_s = \frac{4\pi f_e}{p} = 314.16 \text{ rad/s}$$

The rotor impedance referred to the stator at $s = 0.05$ is:

$$Z_2 = \frac{R_2}{s} + jX_2 = 1.36 + j0.18 \Omega$$

The parallel branch impedance (magnetizing reactance and rotor impedance):

$$Z_p = \frac{jX_M \cdot Z_2}{jX_M + Z_2} \approx 1.258 + j0.404 \Omega$$

Total equivalent impedance:

$$Z_{eq} = R_1 + jX_1 + Z_p = 1.336 + j0.584 \Omega$$

$$Z_{eq} \approx 1.458 \angle 23.6^\circ \Omega$$

The line current (equal to phase current in Y-connection):

$$\mathbf{I}_1 = \frac{V_\phi}{Z_{eq}} = \frac{230.94 \angle 0^\circ}{1.458 \angle 23.6^\circ} = 158.4 \angle -23.6^\circ \text{ A}$$

Stator power factor:

$$PF = \cos(23.6^\circ) = 0.916 \text{ lagging}$$

Stator copper loss:

$$P_{scl} = 3I_1^2 R_1 = 3(158.4)^2 (0.078) \approx 5871 \text{ W} = 5.87 \text{ kW}$$

Input power:

$$P_{in} = \sqrt{3} V_L I_L \cos \theta = 100.5 \text{ kW}$$

Air-gap power:

$$P_{ag} = P_{in} - P_{scl} - P_{core} = 93.33 \text{ kW}$$

Converted power (Mechanical power developed):

$$P_{conv} = (1 - s)P_{ag} \approx 88.66 \text{ kW}$$

Output power:

$$P_{out} = P_{conv} - P_{mech} = 87.56 \text{ kW}$$

Electromagnetic torque:

$$T_e = \frac{P_{ag}}{\omega_s} \approx 297.1 \text{ N m}$$

Rotor speed:

$$\omega_m = (1 - s)\omega_s = 298.45 \text{ rad/s}$$

Load torque:

$$T_{load} = \frac{P_{out}}{\omega_m} \approx 293.4 \text{ N m}$$

Overall efficiency:

$$\eta = \frac{P_{out}}{P_{in}} \approx 87.1\%$$

Using the Thevenin equivalent circuit:

$$V_{th} \approx V_\phi \frac{X_M}{X_1 + X_M} \approx 225.07 \text{ V}$$

$$R_{th} \approx R_1 \left(\frac{X_M}{X_1 + X_M} \right)^2 \approx 0.0741 \Omega$$

$$X_{th} \approx X_1 = 0.18 \Omega$$

The slip at maximum (pull-out) torque is:

$$s_{max} = \frac{R_2}{\sqrt{R_{th}^2 + (X_{th} + X_2)^2}} \approx 0.185$$

The pull-out torque is:

$$T_{max} = \frac{3V_{th}^2}{2\omega_s \left[R_{th} + \sqrt{R_{th}^2 + (X_{th} + X_2)^2} \right]}$$

$$T_{max} \approx 547.8 \text{ N m}$$

For maximum torque to occur at starting, we need $s_{max} = 1$. This means the total rotor resistance referred to the stator ($R_{2,new}$) must equal the denominator of the s_{max} formula:

$$R_{2,new} = \sqrt{R_{th}^2 + (X_{th} + X_2)^2} = 0.3675 \Omega$$

The external resistance referred to the stator is:

$$R'_{ext} = R_{2,new} - R_2 = 0.2995 \Omega$$

To find the actual physical external resistance to be added per phase to the rotor, we divide by the square of the turns ratio (a^2):

$$R_{ext} = \frac{R'_{ext}}{a^2} \approx 0.0749 \Omega \text{ per phase}$$

The value of the starting torque with this added resistance is simply the maximum torque computed above:

$$T_{start} = T_{max} \approx 547.8 \text{ N m}$$

3 Problem 3

The synchronous speed in rpm and rad/s is:

$$n_{sync} = \frac{120f}{p} = \frac{120 \times 60}{4} = 1800 \text{ rpm}$$

$$\omega_{sync} = 1800 \times \frac{2\pi}{60} = 188.5 \text{ rad/s}$$

Since the motor is Y-connected, the phase voltage is:

$$V_\phi = \frac{V_L}{\sqrt{3}} = \frac{230}{\sqrt{3}} = 132.79 \text{ V}$$

To simplify the torque equations, we find the Thevenin voltage (V_{th}) and impedance ($Z_{th} = R_{th} + jX_{th}$) looking into the circuit from the rotor terminals.

$$V_{th} \approx V_\phi \left(\frac{X_M}{X_1 + X_M} \right) \approx 128.79 \text{ V}$$

$$R_{th} \approx R_1 \left(\frac{X_M}{X_1 + X_M} \right)^2 \approx 0.320 \Omega$$

$$X_{th} = X_1 \parallel X_M = 0.485 \Omega$$

Since rotational losses are zero, the developed mechanical power (P_{conv}) equals the output power:

$$P_{conv} = P_{out} = 7460 \text{ W}$$

The load torque at full load ($s = 0.04$) is:

$$T_{load} = \frac{P_{conv}}{\omega_m} = \frac{7460}{\omega_{sync}(1-s)} = 41.22 \text{ N m}$$

The torque equation of the induction motor is given by:

$$T = \frac{3V_{th}^2 \left(\frac{R_2}{s} \right)}{\omega_{sync} \left[\left(R_{th} + \frac{R_2}{s} \right)^2 + (X_{th} + X_2)^2 \right]}$$

Let $x = \frac{R_2}{0.04}$. Solving this resulting quadratic equation for x yields $x \approx 5.568$, which gives:

$$R_2 = 0.223 \Omega$$

The slip at which maximum (pullout) torque occurs is:

$$s_{max} = \frac{R_2}{\sqrt{R_{th}^2 + (X_{th} + X_2)^2}}$$

$$s_{max} \approx 0.215$$

The equation for maximum induced torque is independent of R_2 :

$$T_{max} = \frac{3V_{th}^2}{2\omega_{sync} \left[R_{th} + \sqrt{R_{th}^2 + (X_{th} + X_2)^2} \right]}$$

$$T_{max} \approx 97.38 \text{ N m}$$

The starting torque occurs when slip $s = 1.0$:

$$T_{start} = \frac{3V_{th}^2 R_2}{\omega_{sync} [(R_{th} + R_2)^2 + (X_{th} + X_2)^2]}$$

$$T_{start} \approx 46.52 \text{ N m}$$